Effects of Time on Survey Response

SurvMeth 632

Aaron Maitland

September 24, 2024

Outline

Nature of memory for time

- Reconstruction vs. direct retrieval; encoded order info
- Strength assessment of event age
- Precision and effort required for task

Telescoping

- History of the phenomenon
- 3 theories
  - Time compression, Accessibility, Variance theories

Seam Effects

- Forgetting, constant wave responding, or both?

Duration Estimation

- The way an event is experienced affects perception of its duration – including duration of completing a questionnaire

Time of occurrence

Now we would like to ask you a few questions about the training that you were receiving on (DATE OF LAST INTERVIEW) at (TYPE OF TRAINING AGENCY CODED IN Q.6B). First, when did you finish or leave this training program? [RECORD MONTH AND YEAR]

(National Longitudinal Study [illegible] Market Behavior/Youth)

Duration

Altogether, for how many weeks did you attend this training? [RECORD NUMBER OF WEEKS] (NLS/Y)

Intervals

Please tell me each period between (DATE IN Q.6) and (now/DATE in Q.7B), during which you didn't work for this employer for a full week or more. [ANSWERS GIVEN IN EXACT DATES.] (NLS/Y)

Survey Questions Concerning Time

- Time of occurrence
  - e.g., On what date did the burglary occur?
- Intervals
  - e.g., List start and stop times/dates, possibly for > 1 period
- Duration
  - e.g., For how long were you hospitalized?
- Elapsed time
  - e.g., How long has it been since you last smoked a cigarette?
- Frequency questions
  - How many hours per week do you usually work?

Temporal Information (Potentially) Available for Answer

- Exact date (e.g., *January 16, 2006*)
- Relative order (e.g., *after Spring break*)
- Other details supporting reconstruction (e.g., *springtime; Friday; end of the semester)*
- Metamemory (e.g., *I can't remember much so it must have happened a long time ago*)

Memory for time

- Friedman's (1993) review of laboratory and everyday memory studies for time concluded:
  - in most studies requiring temporal judgments about past events, times are *reconstructed* from context
    - other events for which time is known (landmarks)
    - location in well-known pattern like time of day, day of week, week in semester, etc.
    - physical cues like weather, daylight, clothes

Memory for time (2)

- Friedman (1993) (cont'd)
  - dates rarely directly and automatically coded (tagged)
    - Burt (1992) coded verbal reports (think alouds) while people dated events; only 21% involved direct retrieval of a date
    - exception for special events like birth of a child
- Wagenaar (1986) tested own memory over four years
  - Recorded 4 attributes of daily, autobiographical events: what, who, where, when
  - At time of recall, successively revealed cues for each event, e.g., what, then (if event not recalled) who, then (if event not recalled) where, then...
  - Displaying each of the cues helped recall … *except date (when)*
- Although cannot often be directly retrieved, dates often *reconstructed*

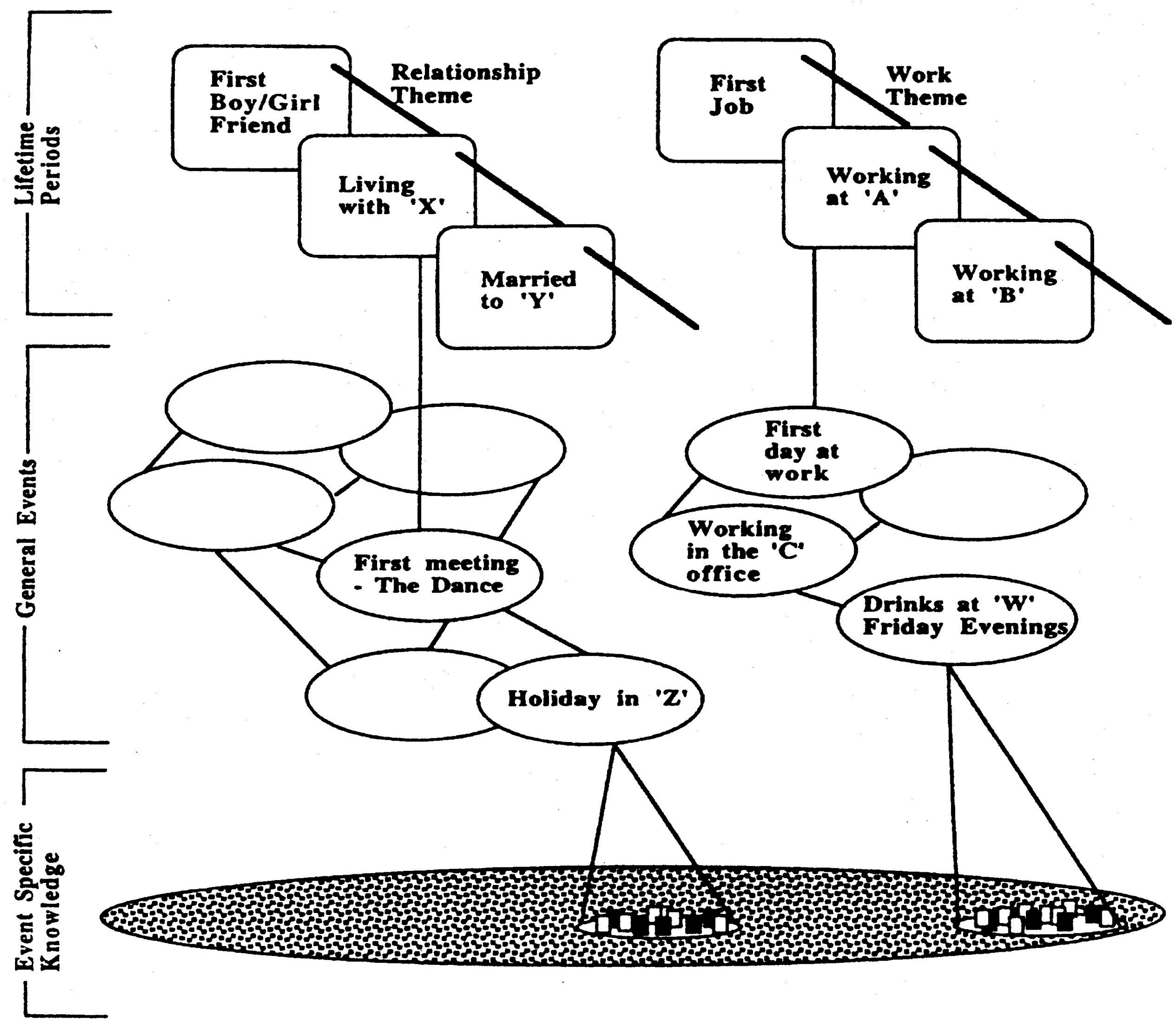
Memory for time (3)

Friedman (cont'd):

- Some evidence that ordered relations (relative time) are encoded with events – the "order code" theory
  - before-after relations coded when later events bring to mind earlier ones
    - Hearing about Chernobyl nuclear accident when it occurred (1986), may have brought to mind similar Three Mile Island nuclear accident (1979), at which point order of the two events coded in people's memories
  - Laboratory evidence:
    - Relative recency judgments (which more recent?) more accurate for words that are related semantically (e.g., *sofa ... chair*) and associatively (e.g., *smoke ... tobacco)*
    - Presumably thinking about one member of a category or association brings to mind related "events" (words) at which point people encode order of the events

Temporal Info in Autobiographical Memory (AM)

- Order info encoded in:
  - *"Lifetime Periods"*
  - *"Autobiographical Period"*
  - various proposals for narrative structure of AM
- Thematic relations in these structures rely on order of events so the "order code" theory could be the mechanism by which order info encoded
- Order-code theory may be one way that survey *R*s reconstruct *when* events occurred


Lifetime Periods
General Events
Event Specific Knowledge
First Boy/Girl Friend
Relationship Theme
Living with 'X'
Married to 'Y'
First Job
Work Theme
Working at 'A'
Working at 'B'
First day at work
Working in the 'C' office
Drinks at 'W' Friday Evenings
First meeting - The Dance
Holiday in 'Z'

Memory for time (4)

- Not much evidence "strength" of stored events used to date them
  - Better memory for *day of week* than year, argues against strength accounts – at least as only account:
    - Day of week is accurately recalled over very long periods of time; seems to be encoded as part of event
    - If strength of memory (distance from present) is primary source of memory for time, day of week should be forgotten at least as rapidly as year
  - "Chronological illusion": memory for time seems like a simple linear sequence… but is not
    - Strength (linear sequence) may be a fallback approach when people have little other information available
  - But some tasks may call for "strength" judgment
    - Did we have Mexican or Chinese food more recently?
    - Which shirt did I wear longer ago?

Memory for Time (5)

- Format of answers affects dating accuracy
- Janssen et al. (2006) asked *P*s to date personal and news (public) events using absolute (e.g., "July 2004") or relative ("3 weeks ago") formats
  - Absolute date assumed to require reconstruction, e.g., from context
  - Relative date assumed to require clear and familiar memories
- *P*s more likely to date *p*ersonal events in absolute (54%) and news events in relative (58%) format
- Events dated more precisely (d/m/y) if personal (63%) than public (35%)
- When given choice, *P*s preferred to date
  1. news events in relative format and personal events in absolute format
  2. remote (older) events in relative format, recent events absolute format

Telescoping

- In answering survey questions with time ("reference") periods, *R*s erroneously include events from outside the reference period
  - (Forward) Telescoping: tendency to report events that occurred *before* the reference period as having occurred *within* it
  - Backward Telescoping: tendency to report events that occurred *after* reference period as having occurred *within* it
    - Only possible if reference period does not "touch" the present
  - Forward displacement more common and usually larger than backward displacement
    - Cannot displace events from future, i.e., that haven't yet occurred, into reference period
  - Tipping point between forward and backward telescoping ~1000 days according to Janssen et al. 2006

Telescoping (2)

- Visual metaphor for subjective temporal distance
  - elapsed time seems to shrink the way physical distance seems to shrink when viewed through a telescope
- Used to account for *overestimates* in frequency reports
  - but may be (more than) offset by forgetting across all *R*s

Telescoping (3)

- Neter & Waksberg (1964) observed misreporting in longitudinal survey due to dating error
  - *R*s report more expenditures for period since last interview when not reminded of what they reported in last interview (unbounded) than when they were reminded (bounded)
  - assumed to reflect forward temporal displacement of events into most recent interval, thus "re-reporting" of events
  - Bounding ("since the last interview…") filters out previously reported events
  - "Pile-up" near the present
    - events cannot be displaced from the past into the future

Telescoping (4)

- Neter & Waksberg measured telescoping within one interview
  - one group asked about each of the previous 3 months
  - second group asked only about last month
  - "last month only" group reported more expenditures for <u>last month</u> than did "previous 3 months" group
  - assumed to reflect forward telescoping, i.e., events temporally displaced into what seemed like last month
- Sudman, Finn & Lannom (1984) bounded recall within the interview
  - Asked about events 30-60 days ago, then 0-30 days ago
  - discarded earlier data since likely included telescoped events

1. Telescoping as memory illusion: time seems to be compressed

- First formal account of telescoping: Sudman & Bradburn (1974)
  - some events are forgotten (leading to underestimation of event frequency)
  - but if events <u>are</u> recalled, they will <u>seem more recent</u> than, in fact, they are
  - "time compression"
  - May lead to events being "imported" into reference period from prior to reference period
  - *time compression* is explicitly represented in model

Telescoping as memory illusion (2)

- Sudman & Bradburn model time compression as *extended subjective reference* period

  Subjective recall period = t + log(bt)
  - t is objective reference period
  - b transforms actual time to subjective time and
- Suppose *R*s asked about behavior over last 3 years and b = 2.5

  Subjective recall period = 3 + log(2.5*3)=3.87
  - this is 29% longer than actual recall period (1.29 x 3 = 3.87) so they assume 29% of events reported in reference period occurred before start of reference period

2. Telescoping and Knowledge

- Subjective event age is shifted based on what *R*s know about events (Brown, Rips & Shevell, 1985)
  – Accessibility principle:
    - the more that is known about an event, the more recent it seems; the less that is known, the less recent it seems
    - people use the amount of knowledge as indication of event age
      – similar to availability heuristic (week 6)

Telescoping and Knowledge (2)

- *P*s date public events (newspaper headlines) and rated how much they knew about events
  - Pairs of events selected to be similar in objective age but different in how much known about them (based on others' rated knowledge)
  - Shows how *P*s' knowledge affects when event seemed to occur

	Low Knowledge	High Knowledge
Date-Rate	-.32	.12

Dating error in days

- Perfect dating => zero error; negative error scores indicate event assigned earlier-than-actual date and positive error scores indicate event assigned date more recent than actual date

3. Variance theories of telescoping

- It's possible to explain dating error in terms of less precision in memory for older events
- Rubin & Baddeley (1989) make 2 simple assumptions about dating of events:

  (1) error is <u>unbiased</u>

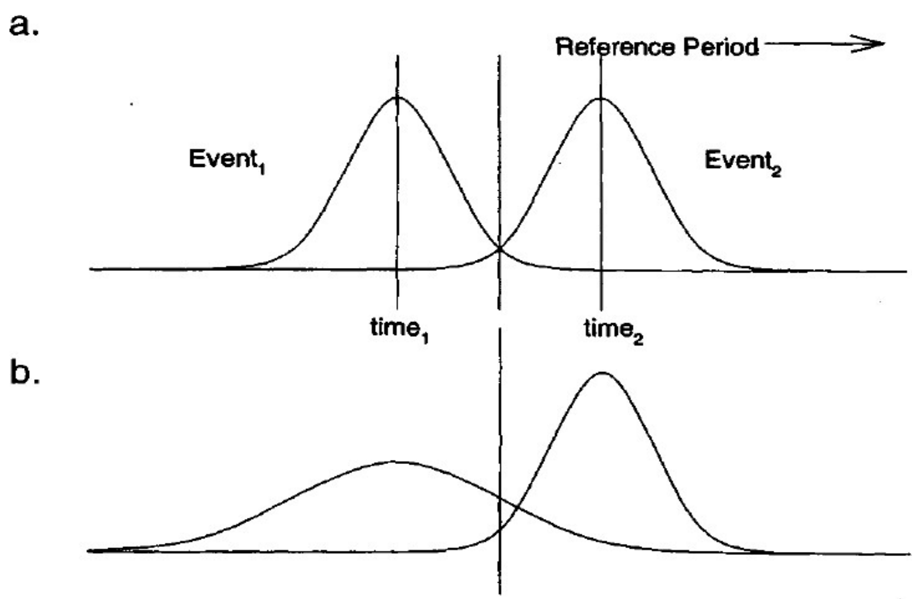
  (2) error is greater for older than more recent events

Unbiased Dating Error

- First assumption predicts equal forward and backward displacement
  1. an event is equally likely to be subjectively moved forward and moved backward in time
  2. dating error is distributed symmetrically around the objective date (variance around the mean)

Greater Dating Error for Older Events

- Second assumption leads to larger forward than backward displacement
  - Dating error should be greater for events that occurred prior to the reference period (older) than events that occurred within reference period (more recent)
  - distribution of dates for older events should have longer tails and lower peaks (means)
  - So, some events from prior to reference period erroneously reported as occurring within reference period ("moved" forward); similar proportion of events erroneously moved back <u>but not reported</u> because seem like they occurred before reference period and in fact did
  - Reporting events from before reference period contributes to measurement error
  - Because error is smaller for newer events, less likely to be moved into reference period when actually occurred after reference period (i.e., between end of reference period and present)
  - displacement within reference period will not be detected
- Result: large forward displacement for older events, small backward displacement for newer events



Source: Tourangeau, Rips & Rasinski (2000)

Area under event$_1$ curve to right of center line represents forward telescoping:
- in (a) equal uncertainty about dates of both events and no net telescoping
- in (b) more uncertainty about date of Event$_1$ which leads to net forward telescoping

Rubin & Baddeley: Empirical Evidence

- Evidence for 2nd assumption (more error for old events):
  - Asked colleagues to recall departmental seminars attended over previous 2 years
    - more recent seminars recalled than older ones
  - provided colleagues with names of colloquia and asked to date
    - Absolute value of dating error increased as age of events increased:

      295, 198, 97, -51, -45

      oldest most recent

Note: negative dating error => subjective dates more recent than actual dates

Rubin & Baddeley

- Evidence for 1st assumption (dating error unbiased):
  - observed small backward displacement when asked about seminars during period ending before present
    - seminars occurring after the reference period reported as having happened within it
- But displacement is greater, even though unbiased, for older events because more memory error for older events

Seam Effect

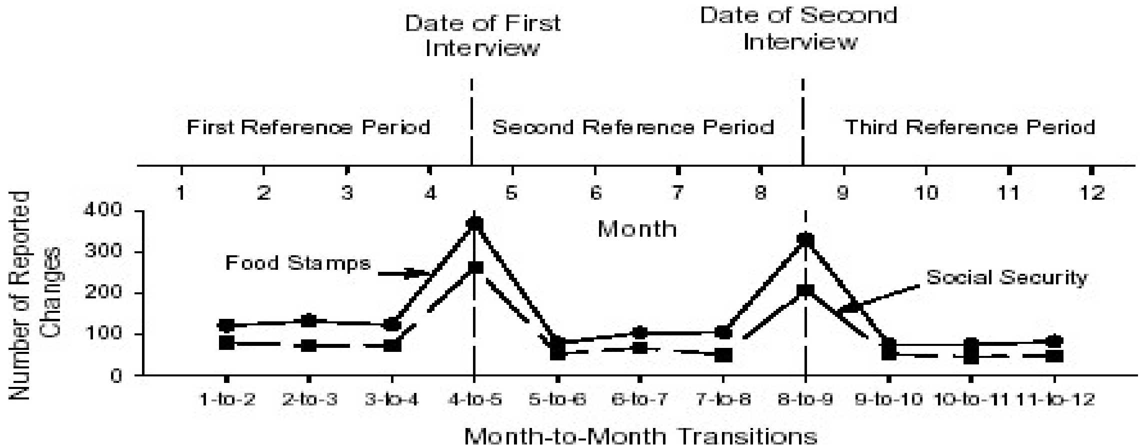
- Memory for the time of events can introduce error in longitudinal (*panel*) surveys like SIPP and PSID
  - where information about several time periods (e.g., months) is collected in a single interview
  - multiple interviews are conducted
  - e.g., interviews conducted in May and Sept about preceding 4 months

Jan Feb Mar Apr May Jun Jul Aug Sept

τ τ

  - Info about April and May collected 4 months apart
- Month-to-month change smaller within than between interviews

Month to month change in Food Stamp and Social Security usage in SIPP



Source: Rips, Conrad & Fricker, 2003

Month to month changes in variables from the Survey of Income and Program Participation (SIPP)

	% change between adjacent months			
Variable	Month 1 to 2 of Ref. Period	Month 2 to 3 of Ref. Period	Month 3 to 4 of Ref. Period	Month 4 of old to Month 1 of new Ref. Period
Marital status	0.3	0.3	0.3	0.7
Employment status	4.6	5.0	5.4	10.2
Personal earnings	5.5	6.3	6.4	16.3
Total family income	4.9	5.4	5.7	17.9
Individual Soc. Security	1.9	1.6	1.8	12.0
Family AFDC receipt	0.1	0.1	0.1	0.3
Family food stamp receipt	0.2	0.4	0.2	0.9

From T, R & R (2000), p.124; source Young (1989)

Seam Effect (3)

- *Forgetting* is certainly part of the seam effect:
  - Memory is better for last month than four months ago
  - Kalton & Miller (1991) found that % of SIPP *R*s who correctly reported actual increase in Social Security benefits in Jan. 1984 is smaller the more time that has elapsed since the event when interviewed
    - February 68.4%
    - March 59.6%
    - April 53.0%
- Exaggerates change across the interviewing seam

Seam Effect (4)

- But *R*s may also minimize changes <u>within</u> reference period
  - *Constant wave responding*
  - Use current value as starting point and adjust (or not) depending on beliefs about stability of behavior or attitude, i.e., did it change?
- Seam effects more likely when memory is poor (Marquis & Moore, 1989)
  - more forgetting and more incentive to use a strategy like constant wave responding

Seam Effect (5)

- Experimental study (Rips, Conrad & Fricker, 2003)
  - *R*s received a mailed questionnaire each week, completed it, and mailed to experimenter
  - The items on each weekly questionnaire became the "events" about which *R* as later asked to report
  - After 4 and 8 weeks, *R* interviewed in lab about questionnaires (events) in previous 4 weeks
    - "In the 4th week's questionnaire, was there an item about taking a day off work due to illness?"

Seam Effect (6)

- If recall fails, constant wave responding more likely
  - Seam effect biggest when memory worst (older events, recall vs. recognition)
  - Constant wave responding most likely when *R*s first answer same question (e.g., purchases at a hardware store) about each week and then another question about each week, etc., than all items about same week (e.g., purchases at hardware store, paid electric bill, bought fast food)

	Blocked	
Recall order	By week	By item
Forward	6.9	21.6
Backward	5.6	16.9

% constant responses

- So, seam effect involves both forgetting (more change across the seam) and constant responding (less change with the seam)

Seam Effect (7)

- Seam effect is reduced by
  - Providing “dependent information”
    - e.g., In the last interview you reported …
    - But does not increase accuracy
  - Blocking by week, i.e., asking about all activities in one period before asking about same activities in other time periods
    - This increased accuracy in one study but not in another
- Not clear that reducing seam effect necessarily reduces error; may just promote smoothing the error across different months

Duration Estimation

- People's ability to estimate durations of events in their lives can affect answers about what they were doing during a time period
  - When *R*s are asked about their time use, e.g., as in American Time Use Survey, must determine how long each event lasted over course of day
  - "I was traveling to the doctor from 8:15 to 8:30 AM" – *R* must have reasoned the travel took about 15 minutes
- How good are people at estimating duration of events?
- How does the character or sentiment of events (positive or negative) impact duration estimates?

Affect and Duration

- Does time fly when we're having fun? Does it drag when we're miserable?
- If so, then measures of time use (e.g., American Time Use Survey) may be distorted by *R*s' affect during reported activities:
  - if *R* feels positive during activity, then shorter duration estimates than if emotion is negative during activity

Affect and Duration (2)

- Freedman, et al. (2014) tested these predictions in a time use study
  - *R*s list all activities and durations between 4AM yesterday and today
  - For three randomly selected activities, *R*s rated how intensely they felt five emotions and two somatic symptoms:
    - Emotions: calm, happy, sad, frustrated, and worried
    - Somatic Symptoms: tired and pain
  - Negative affect (frustrated and tired) positively correlated with duration
    - 18-minute difference between duration of events experienced with no “frustration” (65 mins) and those with frustration (83 mins); 8-minute difference for “tired”
    - Positive emotions negatively related to duration, but effect is eliminated after covariates (e.g., type of activity) are introduced
  - Findings suggest time doesn’t fly when we’re having fun but does drag when we’re feeling bad

Affect and Perceived Speed of Time

- London and Monello (1974) modified speed of an ordinary wall clock while subjects wrote short essays
  - Clock either moved more slowly or quickly than real time
- Speed with which time appeared to move affected judgments of task's interestingness/boringness
  - When clock moved more slowly than actual time, participants rated the task as more boring than when clock moved faster than actual time

Expectations Can Affect Duration Perception

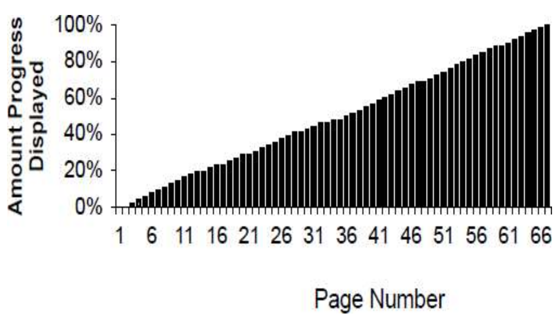
- Boltz (1993) told participants they were one third, halfway, or two thirds finished with a task when in fact they were halfway finished
  - So, *P*s expected remainder of task to last more than, the same as, or less than actual remaining time
- When task lasted longer than expected, *P*s judged it to take longer than it actually did and seemed more fatigued (e.g., longer RT)
- Conclusion: people allocate mental resources on basis of expected duration and, if last longer than expected, “run out of gas”

❖ Suggests *R*s’ expectations about duration of questionnaire could affect completion rates

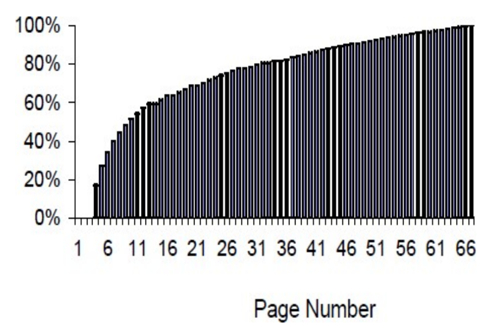
Estimating Duration of Questionnaire: Progress Indicators (*PI*s) in Web Surveys

- Manipulating how *PI* calculates progress (displayed as % at top of page) so that appears to move at different speeds, *R*s' experience differs accordingly
- Conrad, Couper, Tourangeau & Peytchev (2010) displayed progress at 3 speeds:
  - Constant: "50% complete" displayed on page 33 of 67
  - Fast-to-Slow: "50% complete" displayed on page 6 of 67
  - Slow-to-Fast: "50% complete" displayed on page 61 of 67

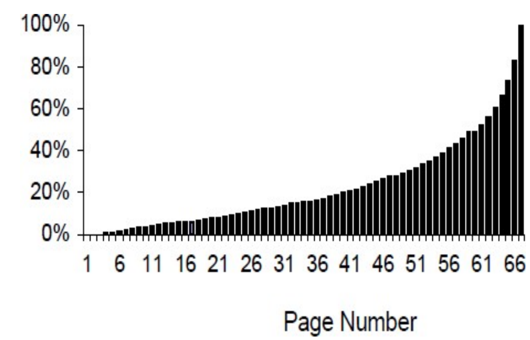
Constant Speed Progress



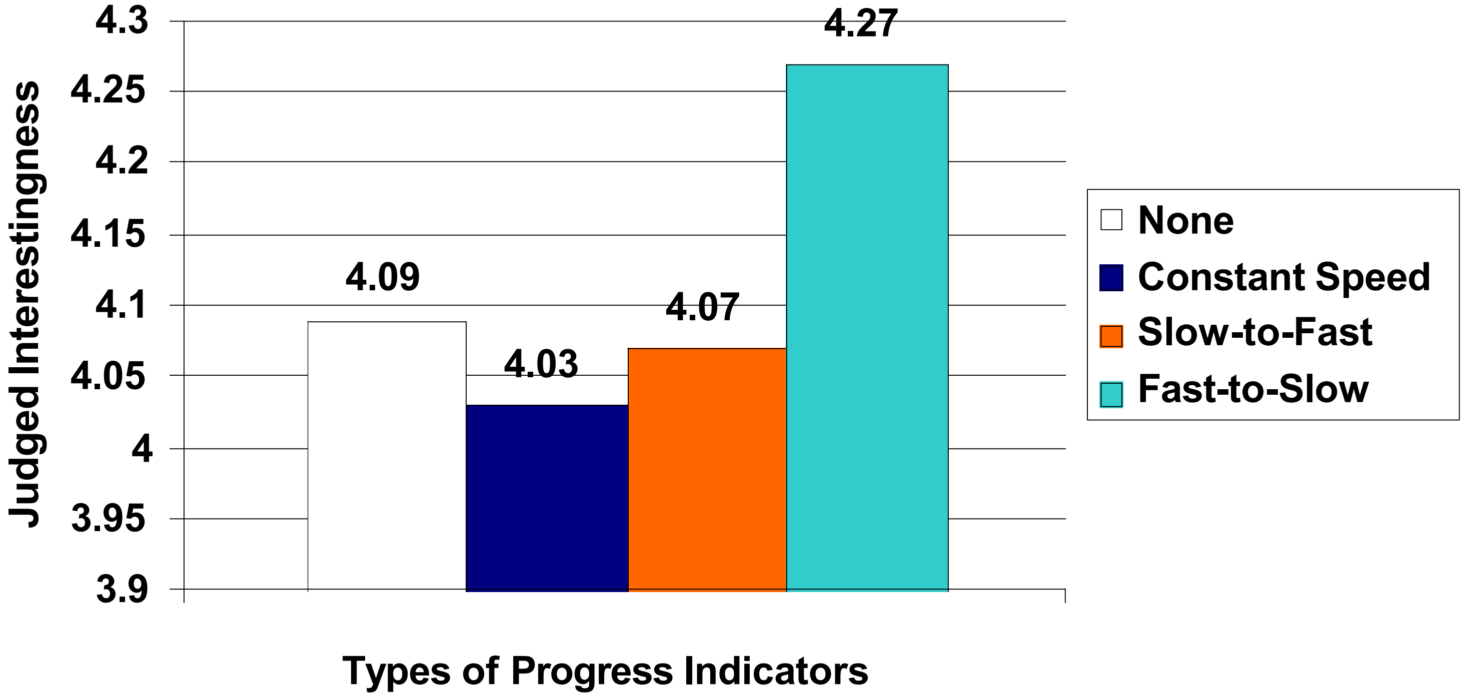
Fast-to-Slow Progress



Slow-to-Fast Progress

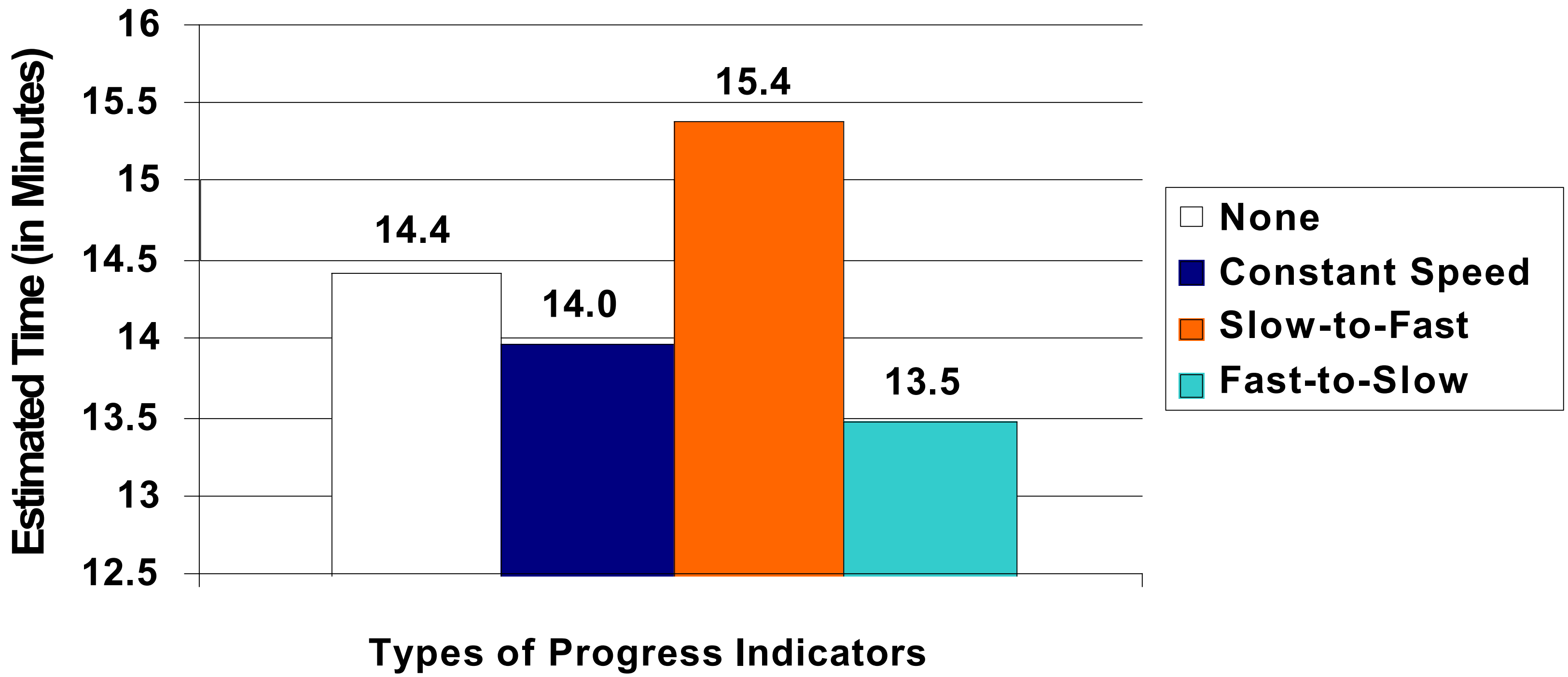


How interesting was the survey?



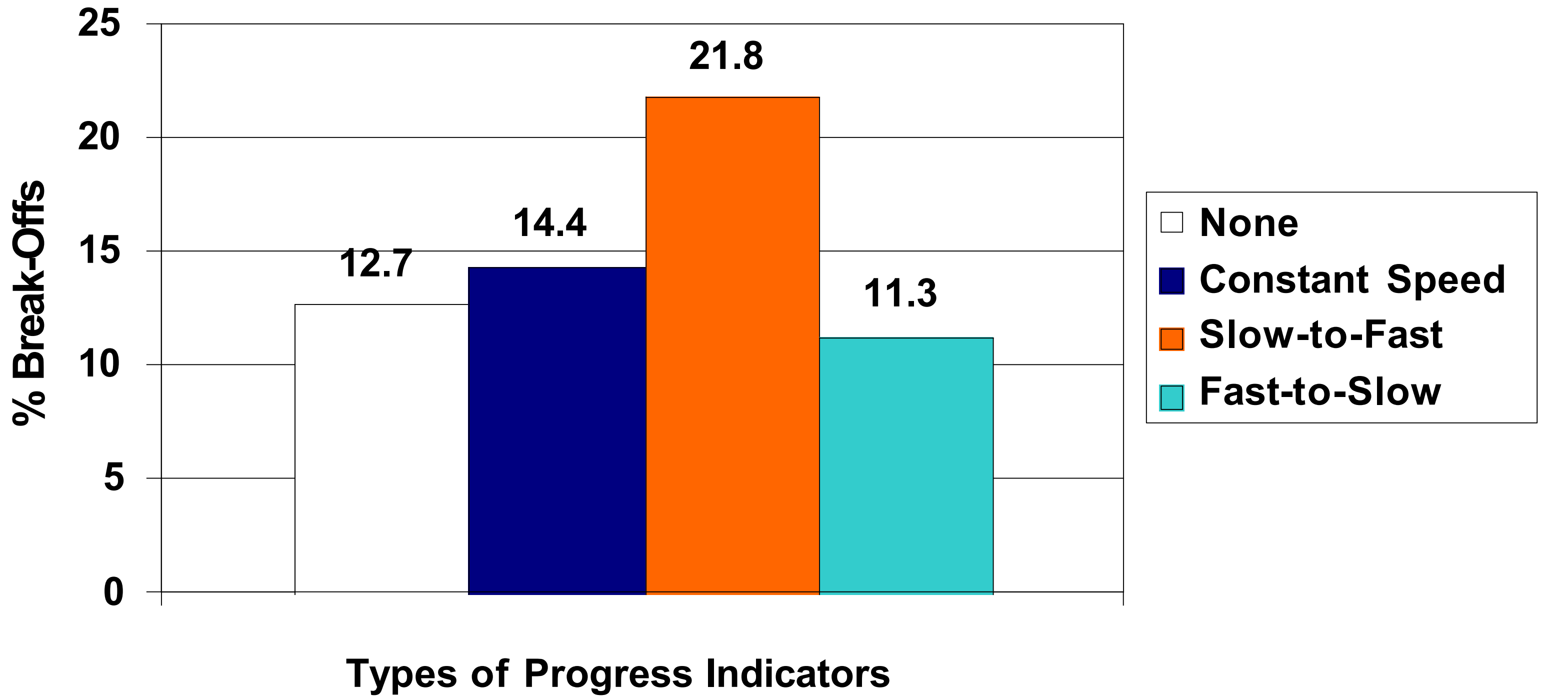
Note: 1= not at all interesting, 5 = extremely interesting; Judgments from Rs who completed questionnaire

How many minutes do you think it took to complete this survey?



Note: Judgments from Rs who completed questionnaire

Effect of *PI* on Breakoff Rates



Summary

- People seem not to encode exact date information but reconstruct it
  - Order of events within narratives usually encoded
- Several accounts of telescoping but, in all, there is both forward and backward subjective movement
  - Backward displacement is smaller
- Seam effects due to both forgetting and constant wave strategy; their reduction does not necessarily reduce measurement error
- Perceptions of how long events take seem related to how people feel during events
  - Can introduce error in measurement of time use
  - Breakoffs can increase when completion time feels longer than expected